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"Knowledge of truth is always more than theoretical and intellectual. It is the product of activity as well as its cause. Scholarly reflection therefore must grow out of real problems, and not be the mere invention of professional scholars."

John Dewey, University of Vermont

Integrating Forecasting and Inventory Decisions Using Machine Learning

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PREVIEW *Can inventory ordering decisions be improved by integrating forecasting and inventory decisions using machine learning? That is the question addressed in this study of three large Belgian companies in the food industry. Van der Haar, Sagaert, and Boute investigate the performance of methods that predict optimal order quantities directly, instead of first forecasting and then calculating optimal inventory quantities. Their results show that using an integrated approach can lead to substantial cost savings for smoother time series, yet the opposite holds when applying it to erratic and lumpy time series.*

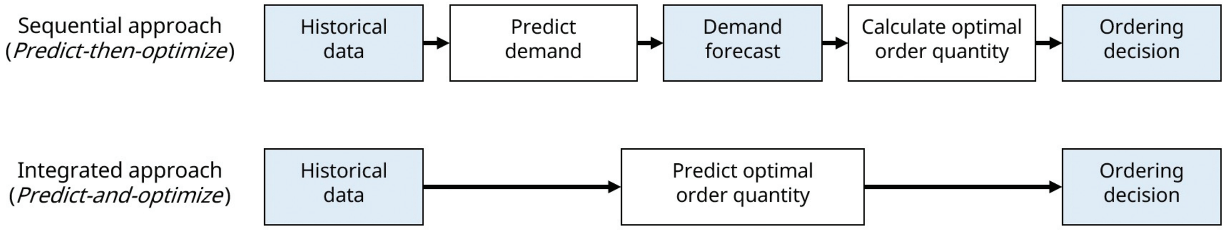
A forecast is only as good as the decision it informs. Good demand forecasts allow organizations to maximize their customer service levels while minimizing inventory-related costs. Defining what makes a good forecast, however, is tricky. Demand forecasting traditionally focuses on predicting demand with maximum accuracy and uses these predictions to inform inventory ordering decisions. Recent research suggests that it may be better to instead directly predict optimal inventory ordering decisions, which maximize service levels and minimize costs. **Figure 1** visualizes the difference between these two approaches.

To understand when an integrated forecasting and inventory optimization approach works best, we first need to understand where sequential approaches fail. Sequential approaches, also referred to as predict-then-optimize approaches, typically involve creating a forecast and assuming a probability distribution on the forecast errors. For example, we can assume that the forecast errors are independent and identically normally distributed. If the assumed distribution fits well, predict-then-optimize approaches can derive replenishment policies that perform well. If not, it may be better to use an integrated predict-and-optimize approach that does not require assumptions on forecast errors.

To see where such assumptions can break down, consider the following examples:

- *Independence:* A series of warmer days in a row can lead to consecutive under- or overestimation of demand, leading to correlation between forecast errors.
- *Identically distributed:* Variance of errors tends to be higher at demand peaks, for example when discounts are offered on products. As a result, errors are often not identically distributed.
- *Choice of distribution:* In practice, forecast errors rarely follow a nicely defined probability distribution such as the normal distribution.

Figure 1. Sequential vs. Integrated Approach to Forecasting and Inventory Control



Some of these assumptions can be avoided when using predict-then-optimize methods. For example, the assumption that the forecast error follows a specific probability distribution can be avoided by using the empirical error distribution. Quantile regression can be used to avoid both this assumption and the assumption that forecast errors are identically distributed. Eliminating the independence assumption, however, is more challenging.

In this study, we show when predict-and-optimize methods work well, and when predict-then-optimize methods prevail. To this end, we use historical sales datasets from three large Belgian food companies to compare the performance of several predict-then-optimize models to that of a predict-and-optimize machine learning method. We proposed the latter method in van der Haar et al. (2024), which builds on related works by Ban and Rudin (2019) and Huber et al. (2019), among others.

Our results show that the choice of method can have an enormous impact on the bottom line. The predict-and-optimize method can lead to cost savings of more than 60% in the best case, but it can also lead to cost increases of at most 43% in the worst case. We find that a predict-and-optimize method works best for smoother time series, whereas predict-then-optimize methods perform better for more erratic and lumpy time series.

Key Points

- This study compares the performance of different methods for forecasting and inventory control on data from three large Belgian food companies.
- Better forecasts do not necessarily lead to better inventory decisions. Correctly identifying the distribution of forecast errors is at least as important as minimizing the errors when the goal is to make inventory decisions.
- It may be better to directly forecast optimal order quantities ("predict *and* optimize") instead of first forecasting demand and then determining order quantities ("predict *then* optimize").
- Predict-and-optimize methods can lead to substantial cost savings for smoother time series, whereas predict-then-optimize methods achieve the best results for erratic and lumpy time series.

APPROACH

Whenever you train a forecasting model using supervised learning, you identify a set of forecasting model parameter values such that the model's predictions minimize some error metric over the training data. For example, most models are trained to minimize the mean squared error (MSE) of the model predictions. However, when we optimize our predict-and-optimize models on the cost-based loss metric described in van der Haar et al. (2024), the loss function measures all current and future costs that follow directly from the inventory ordering decision. For example, if the model overestimates demand, we do not measure by how much demand is overestimated, but instead measure the amount of costs incurred as an effect of this overestimation (e.g., holding costs in terms of cost of capital, and perishing costs in terms of raw material costs).

To provide some intuition on what this means, we compare the cost-based loss function to the MSE and the pinball loss in **Figure 2**. The MSE penalizes large errors much more than small errors, and has the nice property that it leads to the maximum likelihood model if errors are normally distributed. When a forecaster wants to obtain quantile forecasts, they can use the pinball loss, which asymmetrically penalizes errors. For example, to obtain a 90% quantile forecast, this function penalizes underestimation nine times as much as overestimation. The cost-based loss is similar to the pinball loss, but penalizes based on prediction outcomes instead of prediction accuracy. For example, if losing sales is nine times as costly as having leftover inventory, this function penalizes ordering too little nine times as much as ordering too much.

Figure 2. Comparison of the Mean Squared Error, Pinball Loss and Cost-Based Loss

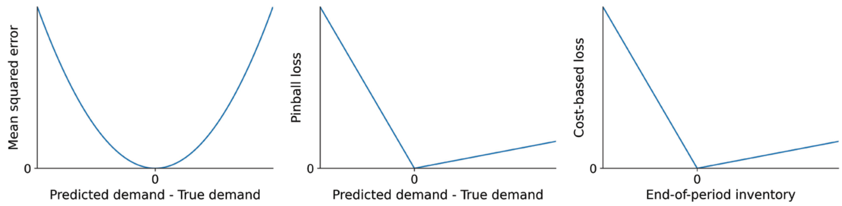


Figure 3. Example of Cost-Based Loss for Perishable Goods

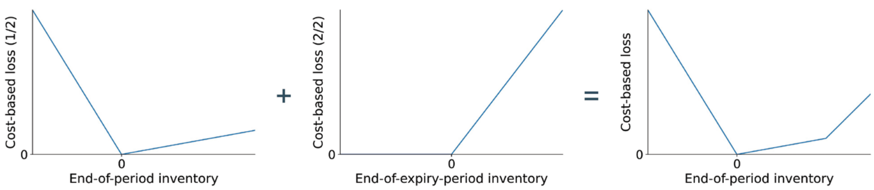


Figure 2 shows what the pinball loss and cost-based loss may look like when aiming for a 90% service level.

Cost-based loss, however, is much more flexible than pinball loss. The cost structure visualized in Figure 2 makes sense when dealing with nonperishable goods, where we trade off a risk between having too much or too little in the period for which we're ordering. For perishable goods, we might also want to look at how much of what we order is left at its expiry date, such that we can account for expiry-related costs. Cost-based loss allows us to do exactly this, as visualized in **Figure 3**. This figure shows that the cost-based loss for nonperishable goods (left panel) can be augmented with a term that penalizes ordering so much inventory that it leads to products expiring (middle panel). The resulting function (right panel) consists of three parts: The leftmost part gives a linear penalty for each unmet demand. The middle part penalizes leftover inventory that does not lead to perishing, e.g., to account for refrigeration costs. The right part penalizes leftover inventory that is bound to perish.

Table 1. Summary of Data Characteristics for the Different Companies and Product Groups

Company/Category	C1	C2	C3a	C3b	C3c	C3d
ADI	1.00	1.00	1.25	1.00	1.08	1.00
CV2	0.45	0.14	1.48	0.60	0.87	0.57
Products	17	2	3	3	3	3
Data rows	5,596	372	4,383	4,383	4,383	4,383

CASE STUDIES

We evaluate the performance of this predict-and-optimize approach on proprietary data from three large Belgian companies in the food industry. These were a ready-made meal company, a catering company, and a supermarket. Company C1 has 10 years of weekly sales records and data on ingredients and materials contained in their products, company C2 has 3.5 years of weekly sales data, and company C3 has 4 years of daily sales data on products grouped into four distinct product categories (a, b, c, and d). Product categories a and b involve nonperishable (frozen) products, whereas categories c and d involve perishable products. **Table 1** provides a brief overview of the different datasets in terms of average demand interval (ADI), squared coefficient of variation (CV2), number of products, and number of data rows.

To compare the performance of the predict-and-optimize against the sequential predict-then-optimize methods, we simulate how the inventory systems of the different companies would have performed under each of the methods. For nonperishable goods, we track holding costs for leftover inventory at the end of each period, and penalty cost for each sale that was lost because no inventory was available to meet demand. For perishable goods, we also track how much inventory was lost due to expiries. To obtain a realistic ratio between these costs, we set the holding cost at 1, the perishing cost at 8 and vary over the lost sales cost "p" to simulate the effect of different service levels. Products in category C3c expire after two days in stock, whereas products in category C3d expire after four days.

We compare the performance of eight different methods on these datasets:

- **ASL:** The predict-and-optimize method discussed in the previous section. We use ASL-LS (Approximate Supervised Learning for Lost Sales) to refer to the approach for nonperishable goods and ASL-PG (ASL for Perishable Goods) to refer to the one for perishable goods. We implement both in LightGBM, but note that our method is model-agnostic and can also be used for other model types such as XGBoost and neural networks.
- **QR:** A predict-and-optimize variation of quantile regression (QR) where a LightGBM model is trained to predict the optimal demand quantile using the pinball loss. The order size is then given by the prediction minus the current stock. Pinball loss coefficients are given by the penalty costs for underpredictions, and by the holding/perishing costs for overpredictions.
- **LGBM:** A predict-then-optimize method where a LightGBM (LGBM) model is used to create a forecast, after which safety stock is added based on the empirical distribution of the residuals (LGBM-E) or under an assumed normal distribution (LGBM-N).
- **ETS:** A predict-then-optimize method where an Error-Trend-Seasonality (ETS) model (Hyndman & Khandakar, 2008) is used to create a forecast, after which safety stock is added based on the empirical distribution of the residuals (ETS-E) or under an assumed normal distribution (ETS-N).
- **LR:** A predict-then-optimize method where a local linear regression model (i.e., a linear regression model specific to a single time series) is used to create a forecast, after which safety stock is added based on the empirical distribution of the residuals (LR-E) or under an assumed normal distribution (LR-N).

For predict-then-optimize methods, both the empirical and the normal distributions are fitted on the forecast residuals from the preceding 52 weeks of data.

For each method except for LR, we obtain cost estimates using cross-validation with prequential train/validation/test splits. We use two years of validation data and two years of test data for C1. For C2, we use eight months of validation data and eight months of test data. Finally, we use one year of validation data and one year of test data for C3. Hyperparameter tuning is performed using Optuna. The test procedure is the same for LR, but no validation runs are performed as LR does not have hyperparameters to tune. Models are retrained at the start of every month for each dataset and each method. An illustration of the prequential train/validation/test procedure for company C1 is given in **Figure 4**.

For ease of interpretation, we present our results in terms of standardized costs in **Figures 5**, **6**, and **7**. Figure 5 provides the results for the nonperishable goods in datasets C1 and C2, Figure 6 for the nonperishable goods in C3a and C3b, and Figure 7 for the perishable goods in C3c and C3d. We obtain these standardized costs by dividing the total costs for a given method by the total costs under ASL for any given combination of dataset and cost of lost sales. Consequently, any standardized cost below 1.00 indicates that a method outperforms ASL, while any standardized cost above 1.00 indicates that a method is outperformed by ASL.

Figure 4. Setup of Our Train/Validation/Test Procedure for Company C1

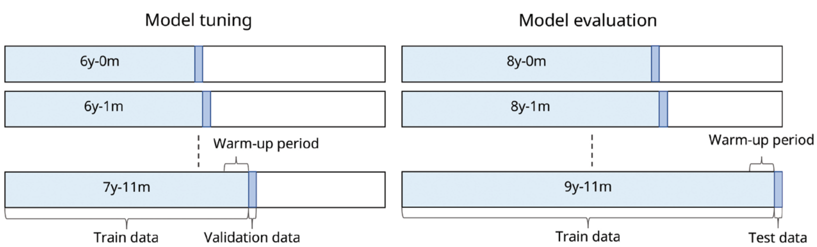


Figure 5. Standardized Costs on Nonperishable Ready-Made Meal and Catering Products

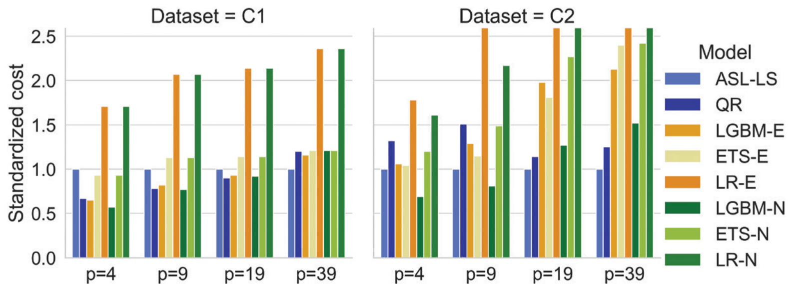


Figure 6. Standardized Costs on Nonperishable Retail Products

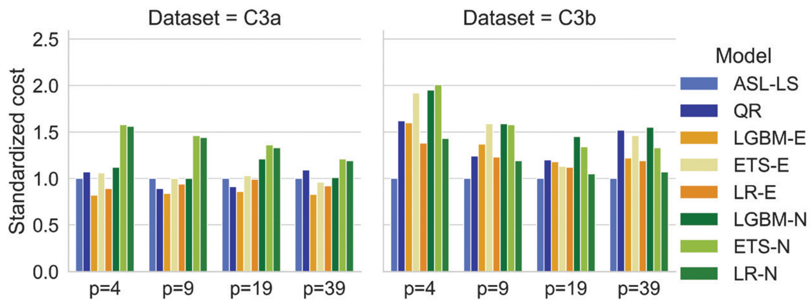
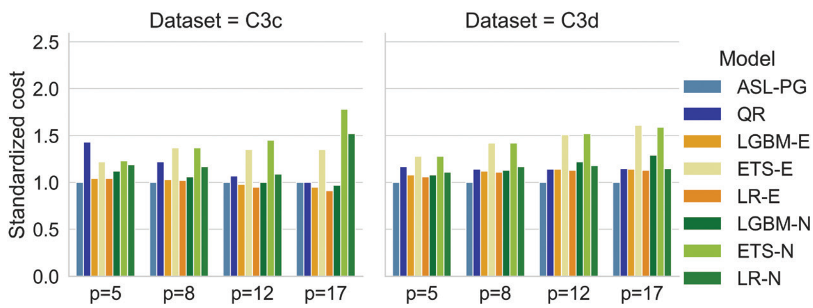


Figure 7. Standardized Costs on Perishable Retail Products



DISCUSSION

The results show that there is no single method that always performs best. We find that the choice for a predict-and-optimize method can lead to either considerable cost savings of more than 60% compared to any benchmark, or considerable cost increases of at most 43%. Cost differences are largest for nonperishable goods (Figures 5 and 6) and appear to be moderated by the penalty p for unmet demand. For companies C1 and C2, the integrated predict-and-optimize method is the best-performing method for the higher service levels (where p is higher). For datasets C3a and C3c, a predict-then-optimize consistently performs on par or best, whereas for datasets C3b and C3d, a predict-and-optimize model consistently performs on par or best.

To understand where performance differences may come from, it is easiest to consider the datasets and their characteristics pairwise. First, both C1 and C2 involve weekly nonintermittent sales data for different nonperishable food offerings with relatively low variation in sales, as measured by the CV2. Second, C3a and C3b contain daily sales data for different nonperishable product categories sold by the same supermarket. Third, C3c and C3d contain the same type of data for different perishable product categories. Note that for each pair, performance of the predict-and-optimize approach is best for the dataset with the lower CV2.

More generally, ASL performs best for 75% of cases where $CV^2<0.7$ (datasets C1, C2, C3b and C3d) and is outperformed for 75% of cases where $CV^2\geq0.7$ (datasets C3a and C3c). For the case with smooth demand where $CV^2<0.7$, ASL leads to average cost savings of 18% compared to the best-performing benchmark (QR). In contrast, for the case with more erratic or lumpy demand where $CV^2\geq0.7$, it would lead to an average cost increase of 8% compared to the best-performing benchmark (LGBM-E).

CONCLUSION

The quality of a forecast depends on more than its accuracy. The ability to accurately describe the distribution of its errors is oftentimes at least as important, as the decisions a forecast informs can be just as dependent on the error distribution as on the forecast accuracy. When the assumed distribution of the forecast errors (e.g., the normal distribution) differs from the true distribution, methods that integrate forecasting and inventory optimization tend to perform well, as they do not involve forecast errors.

The key idea of integrated methods, also known as predict-and-optimize methods, is to directly forecast the optimal order decision instead of first creating a demand forecast and then using it to determine the order decision. We investigate the performance of one such integrated method and find that it can substantially outperform methods that treat forecasting and inventory optimization as two distinct tasks for smooth time series. In contrast, the latter tend to outperform the investigated predict-and-optimize method for more erratic and lumpy time series. We thus conclude that a predict-and-optimize method can aid companies dealing with smoother demand patterns in simultaneously reducing costs and maximizing product availability.

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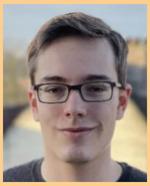

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